Homework 9

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Problem 1. Explain how a TM M' can create a description of the TM M" described above. Remember that M' knows < M > and w before it creates the description of M

Proof. .
We want M" to check if the input is w or not, and if input is w then simulate it on M. If the input is not w then reject. **M' can create this as follows:**

We can use a UTM M' to do this.
We know that multiple tapes are equivalent to a single tape, so lets assume we have multiple tapes. M' adds a copy of w on the tape for M".
M' can also have a copy of the description of M, which it can use to simulate M, since M' is a UTM. This is also added to the tape for M" thus M' can refer to this the transition rules for M.

We can compare the input w", with the w already on the M' tape.
If it doesn't match, then M" rejects the input.
If it matches, then we can simulate the input on M, since M' is a UTM, and it has a description of M. If M accepts the input, then M" will accept. .
Thus, M' can create a description of M" and do everything it does.

□

Problem 2. What does it mean if L(M") is empty, and what does it mean if L(M") is not empty?

Proof. .

L(M") is empty

Suppose L(M") is empty. That means that M" rejects every input string w" to M". Now since M" rejects every input string w" that is not equal to w, and M" simulates M on input w when the input to M" is w (i.e., when w" = w), if M" reject every input string, it must be that M rejects w.

L(M") is not empty

Suppose L(M") is not empty. This means that M" accepts at least one input string w" to M". Now since M" accepts at least one string, and M" can only accept w, since it rejects input of any other form, and it accepts the input after simulating w" on M, it must be that M accepts w.

□

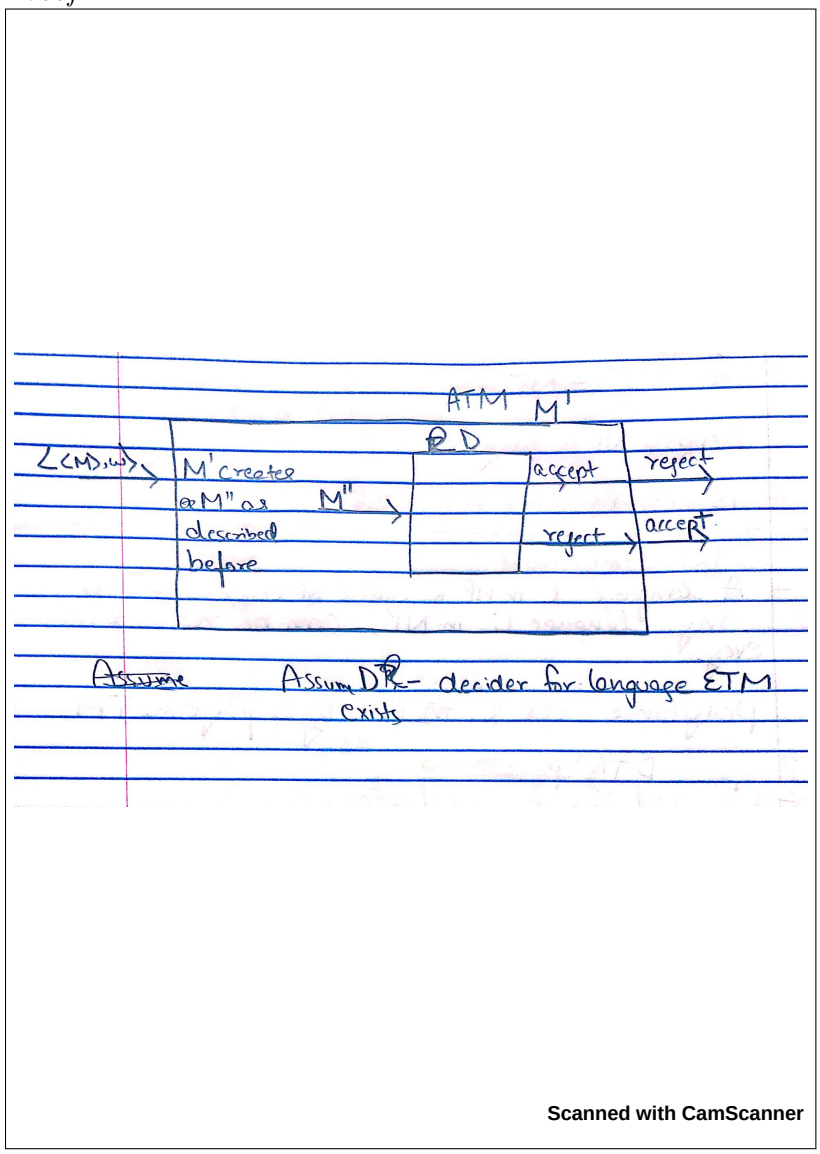
Problem 3. Explain why L(M") is empty if and only if M rejects w,and L(M") is not empty if and only if M accepts w.

Proof. **L(M") is empty if and only if M rejects w,**
If L(M") is empty, then the either the input is not w, and if the input is w, then on simulating w on M, M rejects w.
Thus, if L(M") is empty, M rejects w.
If M rejects w, then w will not be accepted by M", since the way M" is constructed, it can only accept if the input is w and M accepts w, thus M" cannot accept anything, and L(M") is empty.
Thus, if M rejects w, then L(M") is empty.
Thus, proved that if L(M") is empty if and only if M rejects w.

L(M") is not empty if and only if M accepts w,
If L(M") is not empty, then L(M") accepts at least one input string, and the way M" is constructed, it can only the input if the input is w, and on simulating the input string on M, M accepts w.
Thus, if L(M") is not empty, then M accepts w.
If M accepts w, then w will also be accepted by M" since the input will be of the form w, and M" will simulate the input on M, and we know that M accepts w, hence M" accepts w.
Thus, if M accepts w, then M" accepts w.
Thus proved that if L(M") is not empty if and only if M accepts w. □

Problem 4. Complete the proof that M is a TM that can decide ATM.

Proof. .



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We have established the following in the previous question.
1. L(M") is empty if and only if M rejects w.
2. L(M") is not empty if and only if M accepts w.
ATM is a language that given an input of the form $<<M>, w>$ decides if M accepts or rejects w.
ETM is a language that given an input of the form $<M>$, decides if the language M is empty.
We can reduce the input $<<M>, w>$ to $<M>$ i.e. M".
If we have a decider D for language ETM, then we can run M" through D, and if D accepts M" it means that L(M") is empty, and we have established that L(M") is empty if and only if M rejects w. Thus, we know that M rejects w.
If we have a decider D for language ETM, then we can run M" through D, and if D rejects

M" it means that L(M") is not empty, and we have established that L(M") is not empty if and only if M accepts w. Thus, we know that M accepts w.
Thus, we know when M accepts w, and when M rejects w.
Thus, we can decide for the language ATM if we have a decider for ETM.

□

Problem 5. Finish the proof that the language ETM is undecidable.

Proof. If we have a decider for ETM, we can decide ATM.
But, we have established that ATM is undecidable in the previous homework.
Therefore, if ATM is undecidable, and a decider for ETM decides ATM, it must be that the decider for ETM cannot exist since the decider would decide ATM.
Thus, the decider for ETM cannot exist.

□

Problem 6. Look at Sipsers proof that ETM is undecidable. This is the proof of Theorem 5.2 in both editions. He uses different notation,but the ideas of Sipsers proof and the proof above are the same. He uses notation S, R, M, w, M1, and x. State how each of those terms correspond the notation used in the proof above.

Proof. .
S = M'
R = D
M= M
w=w
M1=M"
x=w"

□